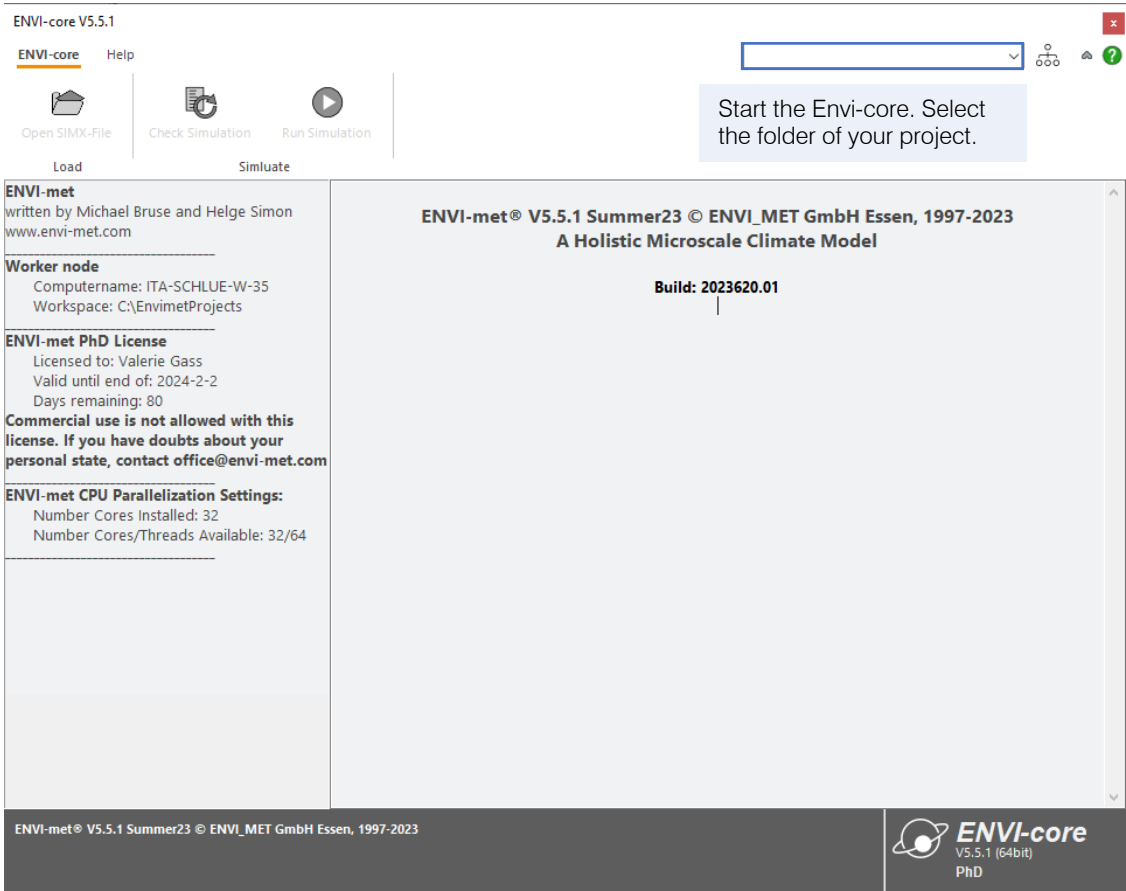
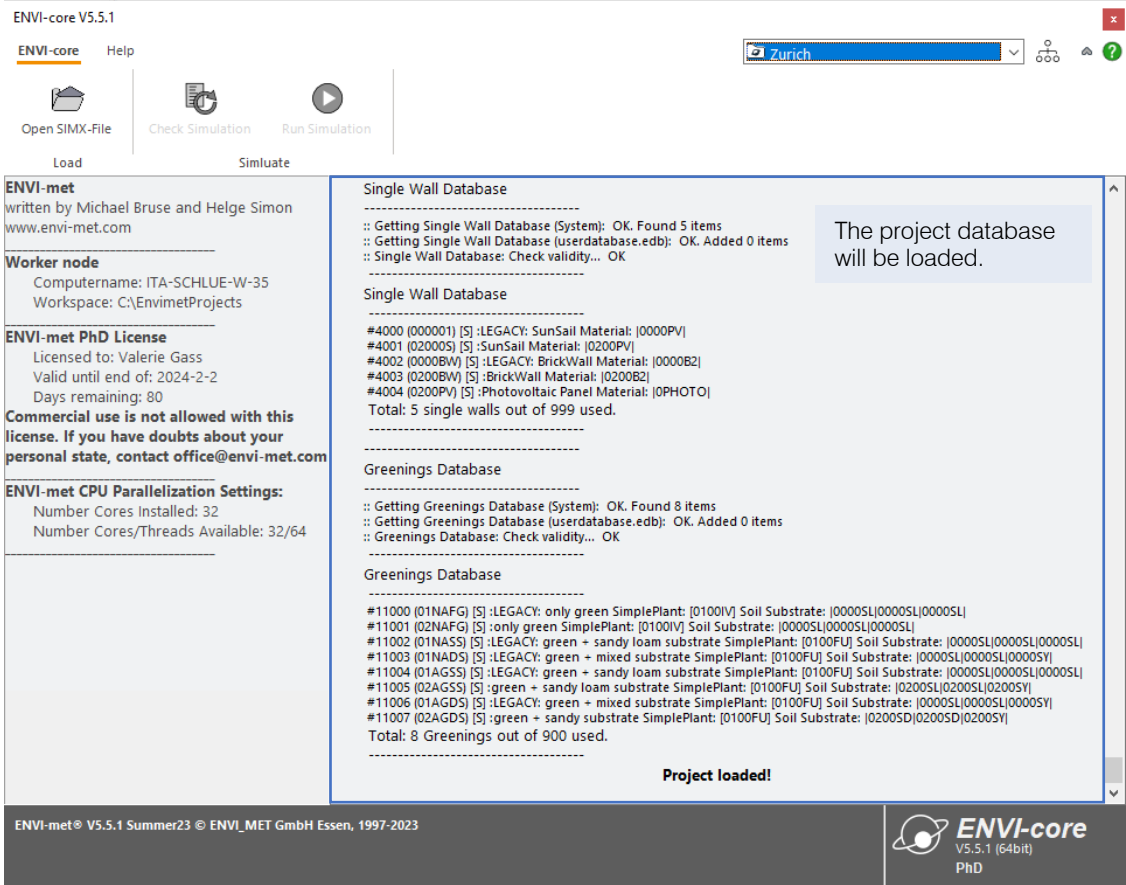


Simulation with Envi-core

1



2



Simulation with Envi-guide

3

ENVI-core: sim_file.SIMX

Open your .SIMX file.

Open SIMX-File | Check Simulation | Run Simulation

Load | Simulate

ENVI-met
written by Michael Bruse and Helge Simon
www.envi-met.com

Worker node
Computernamen: ITA-SCHLUE-W-35
Workspace: C:\EnviMetProjects

ENVI-met PhD License
Licensed to: Valerie Gass
Valid until end of: 2024-2-2
Days remaining: 80

Commercial use is not allowed with this license. If you have doubts about your personal state, contact office@envi-met.com

ENVI-met CPU Parallelization Settings:
Number Cores Installed: 32
Number Cores/Threads Available: 32/64

Project loaded!

```

%% Parsing: "C:\EnviMetProjects\New Project 1\sim_file.SIMX" ...
:: Model Loader: C:\EnviMetProjects\New Project 1\inx_file.INX. Getting file...
:: Model Description: [Add a remark]
:: Model Author: [Enter model author name]
:: Model loader: Getting memory...
:: Model loader: Clear memory...
:: Model loader: Buildings 2D...
:: Model loader: Simple plants (Matrix)...
:: Model loader: 3D Plants...
:: Model loader: Soil Profiles...
:: Model loader: DEM...
:: Model loader: Sources...
:: Model loader: Detailed Building List...
:: Building List: Defines 41 Building(s)
:: Model loader: Full 3D Model...
:: Model loader: Finished, cleaning up...

** Enabled SimpleForcing for T and q. Reading values... **
Reading Radiation Scheme settings...
.... IVS Method: ON
Reading Parallel CPU settings...
..Parallel CPUs/Threads on System : 32/64
..Parallel CPUs requested: 32
..Parallel CPUs used: 32
..Use multiple CPUs: ON
    
```

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ENVI-core
V5.5.1 (64bit)
PhD

4

ENVI-core: sim_file.SIMX

Open SIMX-File | Check Simulation | Run Simulation

Load | Simulate

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Number Cores Installed: 32
Number Cores/Threads Available: 32/64

Check done, Please see output for errors & warnings ! **

Setting up output files...Please wait
Check done.
\$\$ Finished check

ENVI-core
V5.5.1 (64bit)
PhD

Simulation with Envi-guide

5

ENVI-core: sim_file.SIMX

ENVI-core Help

Open SIMX-File

Check Simulation

Run Simulation

Load Simulate

Zurich

9

6

?

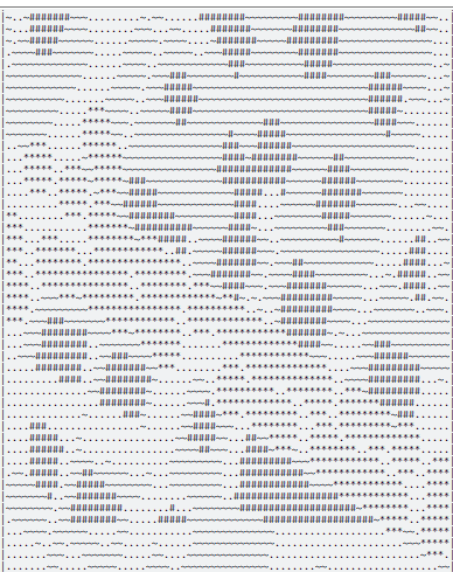
If no errors are observed, click on “Run Simulation”.
The system may not respond while calculating the sky view factors. This process may take some time, depending on your project's size.

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www.envi-met.com

Worker node
Computername: ITA-SCHLUE-W-35
Workspace: C:\EnviMetProjects

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ENVI-met CPU Parallelization Settings:
Number Cores Installed: 32
Number Cores/Threads Available: 32/64



“#” Building, “-” Overhanging Building, “*” Simple Plant “**” 3D Plant “R” Receptor
** Check done, Please see output for errors & warnings ! **

Setting up output files...Please wait
Check done.
\$\$ Finished check

**ENVI-core**
V5.5.1 (64bit)
PhD

6

ENVI-core: sim_file-fullforcing.SIMX

ENVI-core Help

Zurich

9

6

?

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ENVI-met CPU Parallelization Settings:
Number Cores Installed: 32
Number Cores/Threads Available: 32/64

After the first hour of simulation,
an estimated completion time will
be available here.

9%

Simulation will finish around:
2023.11.14 23:46 ±1 hrs

```
07:13:21 24.07.2022
Sun: h=45.29° az=72.46° | Time Step=2.00s | T0ref=28.748°C | q0ref=16.637 | uv(1)=-0.93,0.93 | T(2m)=28.75°C | T(Facade)=32.125°C
max change: T:0.0045K | q:0.0083 | E:0.00005 eps:0.00500 | Km:0.00010 | T_s:0.013°C | Water soil:0.000 | Pollutants:--- | CO2:0.001 | Spray:---
```

ENVI-core
V5.5.1 (64bit)
PhD

```
07:13:21 24.07.2022 ENVI-met 24.07.2022 07:13:21 - 07:13:21
$$ Dynamical time steps:
$$ Sun height <40.00° : dt=2.00 s
$$ 40.00-50.00° : dt=2.00 s
$$ >50.00° : dt=1.00 s

$$ Passive Particle/Gas Concentration:
Not used

$$ Turbulence Closure 3D Mainmodel: TKE
$$ Turbulence Model Version: ENVI-MET/ Bruse 2018 Modification
$$ Turbulence Model LBC: Forced (Closed) Boundary

$$ Starting ENVI-met Mainmodule on 14.11.2023@17:14:08
[0h 18min 53s ] New flow: 1 Steps [101 SOR Steps] du:0.021 dv:0.023 dw:0.028 after 10.10 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)
Dynamical time step adjustment. Set dt=2 sec. (Class 0)
[0h 23min 7s ] New flow: 1 Steps [101 SOR Steps] du:0.019 dv:0.021 dw:0.023 after 10.20 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)
[0h 26min 23s ] New flow: 1 Steps [101 SOR Steps] du:0.017 dv:0.019 dw:0.019 after 10.30 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)
[0h 29min 36s ] New flow: 1 Steps [101 SOR Steps] du:0.016 dv:0.017 dw:0.016 after 10.40 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)

[17:30:50] ENVI-met at 24.07.2022 [06.00.01] Sun: h=28.84° a=73.03° T0ref=300.329 q0ref=15.983
[0h 30min 31s ] Time executed in prognostic flow equation: 10.40 s

[0h 33min 46s ] New flow: 1 Steps [101 SOR Steps] du:0.015 dv:0.016 dw:0.014 after 10.50 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)
[0h 37min 2s ] New flow: 1 Steps [101 SOR Steps] du:0.014 dv:0.015 dw:0.012 after 10.60 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)
[0h 40min 29s ] New flow: 1 Steps [101 SOR Steps] du:0.013 dv:0.014 dw:0.011 after 10.70 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)
[0h 43min 58s ] New flow: 1 Steps [101 SOR Steps] du:0.013 dv:0.013 dw:0.011 after 10.80 s total
Maximum divergence in flow field= 0.70894 @ 76,1,25 (absolute)/76,1,25 (without nesting grids)

[17:45:12] ENVI-met at 24.07.2022 [07.00.01] Sun: h=42.94° a=72.66° T0ref=301.539 q0ref=16.532
[0h 44min 53s ] Time executed in prognostic flow equation: 10.80 s
```